

[illegible]

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

COMMON TO B.E – CSE & M.TECH - CSE

Answer ALL questions

Maximum: 100 Marks

(7x 1 = 7 Marks)

- 24-25-26

7. **CO3** Which section of a Windows executable file is usually targeted by malware to inject code?

- a).data b) .rdata c) .text d).code

Fill in the blanks:

(3x 1 = 3 Marks)

8. **CO1** Wireshark tool is commonly used for _____.
9. **CO2** _____ is a type of cybercrime that involves using email as a medium to send threatening or extortionist messages to a victim.
10. **CO3** _____ refers to the practice of impersonating someone else's email address to mislead the recipient into believing the email is from a trusted source.

PART - B

(5x 3 = 15 Marks)

11. **CO1** Mention the significance of disk imaging forensic investigations.
12. **CO1** List any two tools used for live data acquisition.
13. **CO2** State any two reasons why cookies are important in browser forensics.
14. **CO2** Describe the importance of documentation in evidence acquisition.
15. **CO3** Mention the role of memory dump in malware analysis.

PART - C

(5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO1** i) Explain the standard procedure to be followed during a computer forensic investigation with example. (8)
- CO1** ii) Illustrate the steps to recover erased or damaged data from a hard drive using forensic tools. (7)
17. **CO1** i) Compare and contrast the roles of data encryption and data compression in digital forensics. (8)
- CO1** ii) With suitable examples, describe the process of reviewing network logs. (7)
18. **CO1** i) Explain different types of internet threats with suitable examples. (8)
- CO2** ii) Consider a scenario where a person has been threatened through email. As an investigating officer summarize the steps to extract and interpret the email header to trace the origin. (7)

19. **CO1** i) Explain the process of web browsing activity reconstruction. (8)
CO2 ii) Explain the role of cookies, cache, and temporary files in browser forensics to analyze a suspect's internet activity. (7)
20. **CO2** i) Demonstrate the step-by-step procedure for collecting and analyzing data from a mobile device like smartphone or tablet during investigation. (8)
CO2 ii) Explain the role of the Information Technology Act in handling cybercrime and digital evidence in India. (7)
21. **CO3** i) Summarize the challenges faced by forensic experts when giving evidence in court. (8)
CO3 ii) Discuss various reverse engineering techniques in malware analysis. (7)
22. **CO3** i) Explain the process of extracting malware from a compromised Linux system. (8)
CO3 ii) Discuss various rootkit detection techniques. (7)
